ARYAN BABURAJAN

aryanbaburajan2007@gmail.com | GitHub | LinkedIn

EXPERIENCE

Frappe.io
Jan – Apr 2026

Intern

- Wrote a proof-of-concept Gunicorn worker using Python 3.14 Subinterpreters, enabling true concurrent parallel execution beyond the limitations imposed by the GIL.
- Integrated Bloomberg's Memray profiler into Frappe's web dashboard to support memory and performance analysis.
- Extended Redis Queue's WorkerPool to allow workers to process multiple queues concurrently, improving memory usage by 32%.

PROJECTS

Ducktape
2021 – 2025

Creator

- Built an open-source C++ 3D game engine with an Entity Component System architecture using EnTT, reaching 141+ GitHub stars and 28 forks.
- Implemented real-time rendering in OpenGL, with previous renderer implementations in Vulkan.
- Developed native C++ DLL-based scripting and an ImGui editor for in-engine debugging and content creation.

SerenityOS
Open Source

Contributor

- Improved Unicode support across native applications, extending character handling beyond ASCII.
- Migrated applications to SerenityOS's GML markup language.

Mirage
Personal Project

Creator

- Built a personalized product discovery platform with an authenticated dashboard tailored to users' interests.
- Implemented an email newsletter system that periodically delivers product recommendations using scheduled jobs.
- Built with Next.js, React, MongoDB, Kinde Auth, and Resend.

X-Embed
Personal Project

Creator

- Built a Twitter/X embedding service that generates screenshot-style previews from post URLs.

ACHIEVEMENTS

- **1st Place (₹15K) - Google Developer Groups Physical AI Hackathon**
- **Game Development Workshops** - Held three workshops for 20+, 50+, and 100+ students on game development using p5.js, Unity, and Three.js.

TECHNICAL SKILLS

- **Languages**: C++, C, Rust, Python, TypeScript, JavaScript, C#
- **Web**: React, Next.js, Nest.js, Express.js, Node.js, WebSockets
- **Graphics**: OpenGL, Vulkan, EnTT, Dear ImGui, Unity, Blender
- **Tooling**: Git, CMake, Linux, MySQL, MongoDB, Netlify, Vercel, Heroku

EDUCATION

Christ College of Engineering
2025 – Present

Bachelor of Technology in Computer Science and Engineering

Kerala, India